Name (Last, First):

NEU ID:

Assignment 9

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 87 = 3 + 84 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:
3 points: Correct answer with work shown
2 points: Incorrect answer but attempt shows some understanding (work shown)
1 point: Incorrect answer but an attempt was made (work shown)
0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]
(link to your live google doc)

I. Errors and Power (3 x 3 = 9 pts)

Your local Starbucks is hoping to cut back on the wait time for their customers. The proportion of drive-through customers who wait longer than 3 minutes to get their coffee is $p = 0.76$. In an effort to reduce this, the manager assigns an extra employee to help make the orders in the morning. During the next month, the manager will select an SRS of customer wait times and determine if the proportion who are waiting longer than 3 minutes has decreased.

a) What is the parameter of interest in this problem?

Answer:

b) State the null and alternative hypotheses.

Answer:

c) Describe a Type I and Type II error in this setting and explain the consequences of each.

Answer:

II. Confidence Intervals and Significance Tests (8 x 3 = 24 pts)

1) Is the ratio of males and females that attend a large university even? An SRS of 254 students that attend the college is taken and 145 of them were female. A 99% confidence interval for p = the proportion of females that attend the university is given by (0.491, 0.651)

(a) Use the confidence interval to draw a conclusion about the hypothesis H_0: p = 0.5 vs the alternative H_a: p ≠ 0.5.Be sure to indicate the appropriate significance level.

Answer:

(b) What information is provided by the confidence interval that would not be provided by a test of significance alone?

Answer:

2) In political campaigns, an attack ad ("mudslinging") is an advertisement whose message is designed to personally attack an opposing candidate in order to gain support for the candidate waging the attack. A recent candidate running for governor noticed his support in the polls had slipped, and only 45% of voters now supported him. His campaign ran the attack ad, hoping to boost his voter approval. Two weeks after the ad ran, the poll was run again and it found that out of the SRS of 450 voters, 233 supported the candidate. The 95% confidence interval for the proportion of voters who support him is (0.472, 0.564).

(a) Suppose his pollsters conducted a test of H_0: p = 0.45 vs the alternative H_a: p ≠ 0.45, using α = 0.05. Use the confidence interval to determine whether this test would reject or fail to reject the null hypothesis. Explain your reasoning.

Answer:

(b) Find the P-value for the test described in part (a), and explain what it measures in the context of the problem.

Answer:

3) Netflix relies heavily on viewers when it comes to deciding if they are going to make another season of their shows. A recent article stated that Netflix has said they will only continue to make another season of a show if it has been watched by over 20% of its viewers in the first month of the show coming out. Recently, Marvel's Daredevil was canceled. Let p = the true proportion of viewers who watch the show.

(a) State the null and alternative hypotheses for this test.

Answer:

(b) Describe a Type I and Type II error in this context, and explain the consequences of each type of error.

Answer:

(c) The network determines that for a sample size of 2000, the power of this test at a 5% significance level for H_0: $p = 0.20$ is only 0.39. Explain what the power of the test measures in the context of the problem.

Answer:

(d) Based on your answers to (b) and (c), would $\alpha = 0.10$ or $\alpha = 0.01$ be a better significance level for this test? Explain your choice.

Answer:

III. Confidence Interval for Means (4 x 3 =12 pts)

Ariel is a huge fan of the Milky Ways. She has a "fun-size" bar every night after dinner. She recently began to wonder, after learning about it in AP Statistics, if the claim on the candy was correct. The wrapper says that the candy is 10 oz in weight. Ariel selected an SRS of 30 Milky Way fun size bars and weighed them. She creates a 90% confidence interval for the true mean weight of the candy bars: (8.92, 10.98).

(a) What is the point estimate from this sample?

(b) What is the margin of error?

(c) Interpret the 90% confidence interval in the context of the problem.

(d) Interpret the confidence level of 90% in the context of the problem.

IV. Significant Test for Means (4 x 3 = 12 pts)

Your local television station claims that in a single hour on their station, less than 10 minutes of advertisements will be played. To investigate their claim, you randomly select 10 different hours over the course of a month and record how many minutes an advertisement is playing. Here is the data:

9	10	12	8	9.5	9	10.5	10	8.5	9.5

Does this data provide convincing evidence that the average time of advertisements playing is less than the TV station's claim? Test this at the 5% significance level. **Use the 4 Step Procedure to perform a significant test.**

Answer:

V. ME and Matched Pairs (3 x 3 = 9 pts)

Use the following scenario for questions 1 – 3 ***[Mark your choice in the boxes provided]****.*

Does listening to positive affirmations tapes (for example, "I am in charge of my life") while sleeping decrease depression? Dr. Goodyear wants to test this theory! Dr. Goodyear randomly selected 7 of his patients with mild depression. He gives them a "test" to score their depression on a 100 point scale, where the higher the score, the more clinically depressed you are. He then assigns the 7 patients to listen to an affirmation tape for one month while sleeping. After one month, the patients are given the test again.

	Test Scores						
Person	A	B	C	D	E	F	G
Before	54	57	68	43	42	49	66
After	52	50	67	44	30	38	61

1) Which of the following conditions must be met in order to use a *t*-procedure on these paired data?

(A) The distribution of both before scores and after scores must be approximately Normal.

(B) The distribution of before scores and the distribution of differences (before – after) must be approximately Normal.

(C) Only the distribution of before scores must be approximately Normal.

(D) Only the distribution of differences (before – after) must be approximately Normal.

(E) All three distributions—before, after, and the difference—must be approximately Normal.

2) If μ_d represents the mean different in before tests – after tests, which of the following would be the appropriate hypothesis to test if the positive affirmation tapes have an effect on depression scores?

(A) H_0: $\mu_d > 0$ and H_a: $\mu_d = 0$

(B) H_0: $\mu_d \neq 0$ and H_a: $\mu_d = 0$

(C) H_0: $\mu_d = 0$ and H_a: $\mu_d < 0$

(D) H_0: $\mu_d = 0$ and H_a: $\mu_d \neq 0$

(E) H_0: $\mu_d = 0$ and H_a: $\mu_d > 0$

3) You run a one sample t test on the mean difference in before tests – after tests and obtain a p-value of 0.0155. If you significance level is 0.05, which of the following is correct?

I. We are 95% confident that we can reject the null hypothesis.

II. We have convincing evidence of a difference in depression test scores.

III. The probability of making a Type I error is 0.0155

(A) I and III only

(B) I and II only

(C) II and III only

(D) II only

(E) I, II, and III

VI. Difference in Means (6 x 3 = 18 pts)

Do high school girls and boys differ in how many hours they get per night? At a large high school, 35 girls were randomly selected and they got an average of 6.7 hours of sleep the night before, with a standard deviation of 0.65 hours. 40 boys were randomly selected and they got an average of 6.4 hours of sleep the night before, with a standard deviation of 0.45 hours. Construct a 90% confidence interval and conduct a two-sided significance test at the 10% significance level. Do these two results support each other? Explain.

Answer:

Significance Test Step 1

Significance Test Step 2

Significance Test Step 3

Confidence Interval Calculation

Conclusions and Explanations (6 pts)

THE END